Spatial SQL - Private Preview - FAQ

Last updated: Dec 19, 2024 | Kent Marten [PM] Michael Johns [Product Specialist]

Dec 19th Update:

The Spatial SQL private preview is now available with:

- Spark compute clusters
- Serverless compute clusters
- DBSQL
- DLT Pipelines

There are new ST_* functions supported in DBR 16.1, now including:

- st_startpoint
- st_endpoint
- st_numgeometries
- st_geometryn
- st_flipcoordinates
- st_asewkt
- st_pointn
- st_multi

Spatial SQL expressions are now available in Pyspark with DBR 16.1.

What is new?

There are 60+ spatial expressions supported in Photon-enabled Databricks Runtimes, starting with version 15.3 with the Private Preview enabled.

Function	Input type(s)	Output type
h3_tessellateaswkb	(STRING, INT[, BOOLEAN])/(GEOGRAPHY, INT[, BOOLEAN])/(BINARY, INT[, BOOLEAN])	STRUCT{BIGINT, BOOLEAN, BINARY}
st_area	GEOMETRY / GEOGRAPHY / STRING / BINARY	DOUBLE
st_asbinary / st_aswkb	(GEOMETRY[, STRING]) / (GEOGRAPHY[, STRING])	BINARY
st_asewkb	(GEOMETRY[, STRING]) / (GEOGRAPHY[, STRING])	BINARY

st_asgeojson	(GEOMETRY[, INT]) / (GEOGRAPHY[, INT])	STRING
st_astext / st_aswkt	(GEOMETRY[, INT]) / (GEOGRAPHY[, INT])	STRING
st_buffer	GEOMETRY, DOUBLE	GEOMETRY
st_centroid	GEOMETRY	GEOMETRY
st_contains	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	BOOLEAN
st_convexhull	GEOMETRY / STRING / BINARY	GEOMETRY
st_difference	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	GEOMETRY
st_distance	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	DOUBLE
st_distancesphere	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	DOUBLE
st_distancespheroid	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	DOUBLE
st_envelope	GEOMETRY / STRING / BINARY	GEOMETRY
st_geogarea	STRING / BINARY	DOUBLE
st_geogfromgeojson	STRING	GEOGRAPHY
st_geogfromtext / st_geogfromwkt	STRING	GEOGRAPHY
st_geogfromwkb	BINARY	GEOGRAPHY
st_geoglength	STRING / BINARY	DOUBLE
st_geogperimeter	STRING / BINARY	DOUBLE
st_geohash	GEOMETRY, INT	STRING
st_geometrytype	GEOMETRY / GEOGRAPHY / STRING / BINARY	STRING
st_geomfromgeohash	STRING	GEOMETRY
st_geomfromgeojson	STRING	GEOMETRY
st_geomfromtext / st_geomfromwkt	(STRING[, INT])	GEOMETRY
st_geomfromwkb	(BINARY[, INT])	GEOMETRY

st_geomfromewkb	(BINARY[, INT])	GEOMETRY
st_intersection	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	GEOMETRY
st_intersects	(GEOMETRY,GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	BOOLEAN
st_isempty	GEOMETRY / GEOGRAPHY / STRING / BINARY	BOOLEAN
st_isvalid	GEOMETRY / STRING / BINARY	BOOLEAN
st_length	GEOMETRY / GEOGRAPHY / STRING / BINARY	DOUBLE
st_makeline	ARRAY{GEOMETRY}	GEOMETRY
st_makepolygon	GEOMETRY[, ARRAY{GEOMETRY}]	GEOMETRY
st_ndims	GEOMETRY / GEOGRAPHY / STRING / BINARY	INT
st_npoints	GEOMETRY / GEOGRAPHY / STRING / BINARY	INT
st_perimeter	GEOMETRY / GEOGRAPHY / STRING / BINARY	DOUBLE
st_point	(DOUBLE, DOUBLE[, INT])	GEOMETRY
st_rotate	(GEOMETRY, DOUBLE) / (STRING, DOUBLE) / (BINARY, DOUBLE)	GEOMETRY
st_scale	(GEOMETRY, DOUBLE, DOUBLE[, DOUBLE]) / (STRING, DOUBLE, DOUBLE[, DOUBLE]) / (BINARY, DOUBLE, DOUBLE[, DOUBLE])	GEOMETRY
st_setsrid	(GEOMETRY, INT)	GEOMETRY
st_simplify	(GEOMETRY, DOUBLE) / (STRING, DOUBLE) / (BINARY, DOUBLE)	GEOMETRY
st_srid	GEOMETRY / GEOGRAPHY	INT
st_transform	(GEOMETRY, INT)/(STRING, INT)/(BINARY, INT)	GEOMETRY
st_translate	(GEOMETRY, DOUBLE, DOUBLE[, DOUBLE]) / (STRING, DOUBLE, DOUBLE[, DOUBLE]) / (BINARY, DOUBLE, DOUBLE[, DOUBLE])	GEOMETRY
st_union	(GEOMETRY, GEOMETRY) / (STRING, STRING) / (STRING, BINARY) / (BINARY, STRING) / (BINARY, BINARY)	GEOMETRY
st_union_agg	GEOMETRY / STRING / BINARY (aggregate)	GEOMETRY
st_x	GEOMETRY / STRING / BINARY	DOUBLE
st_xmax	GEOMETRY / STRING / BINARY	DOUBLE
st_xmin	GEOMETRY / STRING / BINARY	DOUBLE

st_y	GEOMETRY / STRING / BINARY	DOUBLE
st_ymax	GEOMETRY / STRING / BINARY	DOUBLE
st_ymin	GEOMETRY / STRING / BINARY	DOUBLE
st_zmax	GEOMETRY / STRING / BINARY	DOUBLE
st_zmin	GEOMETRY / STRING / BINARY	DOUBLE
to_geography	STRING / BINARY	GEOGRAPHY
to_geometry	STRING / BINARY	GEOMETRY
try_to_geography	STRING / BINARY	GEOGRAPHY
try_to_geometry	STRING / BINARY	GEOMETRY

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What geospatial formats are supported?

Most Databricks spatial expressions support reading directly from Well-known Text (WKT), Well-known Binary (WKB), Extended Well-known Binary (EWKB), and GeoJSON. You can also convert to native geospatial types (Geography and Geometry) and use these within a query. Materializing Geography and Geometry values or returning these values in a resultset is not currently supported, you need to convert to WKT, WKB, EWKB, or GeoJSON to do so.

When could I expect to be able to store and return GEOMETRY/GEOGRAPHY columns?

The preview supports in-memory geometry and geography values only which cannot be stored or returned. While we cannot give a timeline, full support for geography and geometry values is on the roadmap to be delivered as the capability further matures.

What if I am already using another lib (e.g DBLabs Mosaic) for spatial processing and analysis?

If you are already using other spatial libraries - you can continue to use them as you do today. This private preview is available with DBR 14.2. However, all external libraries, including DBLabs Mosaic (not to be confused with MosaicML), cannot be executed in DBSQL. If you want to use ST_ expressions in DBSQL, this Preview is the best and only path to do so.

We recommend evaluating the Preview in DBSQL, as well as assessing the suitability of the preview capabilities to replace existing DBR-based workloads.

How do I participate in the Private Preview?

Your account team will need to nominate you first, then you will need to approve the Private Preview Terms of Service to participate. The preview will be enabled by the Databricks Engineering team.

What are the requirements to enable the Preview?

There are requirements for using the Spatial SQL Private Preview:

For DBR:

- Use Databricks Runtime version 14.2+
- Use a Photon-enabled cluster

For DBSQL, Serverless compute, there are no additional requirements.

How do I enable the Private Preview?

Use the Getting Started guide to enable the Private Preview.

What are the limitations of the Private Preview?

- **In-Memory GEOMETRY/GEOGRAPHY Types:** The preview supports in-memory geometry and geography values only which cannot be stored or returned. This requires additional `ST_` calls, such as `ST_AsWKB` or `ST_AsWKT` or `ST_AsEWKB` or `ST_AsGeoJSON`, in writes and queries.
- **Explicit spatial index:** The preview does not automatically perform spatial indexing on GEOMETRY/GEOGRAPHY columns, and spatial joins will not be performant unless customers take explicit actions (described in the Getting Started guide).
- **Map Rendering:** The preview doesn't introduce any additional map rendering support. Beyond what is already available, rendering can be handled by various external libraries and integrations available on the Databricks platform.
- **SQL surface**: GEOGRAPHY and GEOMETRY expressions cannot be used in ORDER BY, GROUP BY or any other clauses that require the underlying type to be comparable or equality comparable.

How is the Private Preview supported by Databricks?

There's no formal support or SLAs for the preview - so please reach out directly to your account team + kent.marten@databricks.com and michael.johns@databricks.com.

What if I find a bug, or identify a potential improvement, in the Private Preview?

We welcome your feedback! Please reach out directly to your account team + kent.marten@databricks.com and michael.johns@databricks.com.

Can I use this in my production environment?

We recognize that the ST_ functions are new for DBSQL, and may unblock certain use cases. However, due to the nature of early previews, we ask you not to put the preview in production until the capability can further mature. We ask you to first evaluate the preview, outside of production, to assess its suitability to meet your specific needs. The Private Preview Terms of Service agreement offers no accommodations for issues arising from using this Private Preview in a production environment.

Can I share information about the Private Preview outside of my organization (including on social media)?

Non-public information about the preview (including the fact that there is a preview for the feature itself) is confidential. As per the Private Preview Terms of Service, you are asked to keep it confidential and do not disclose what you learned in the preview.

Should we decide to release this, we'd love for you to be a reference if you find it useful - please let us know if you'd be willing to do so (your account team + kent.marten@databricks.com and michael.johns@databricks.com).

Thank you for partnering with us in this preview!

-Databricks Team